

T. de perturbaciones a 1 TLS

El hamiltoniano de 1 TLS con forzamiento es $H = \hbar \omega \sigma_+ \sigma_- + \hbar g (\sigma_+ + \sigma_-)$

Y el hamiltoniano efectivo es $K = \hbar \omega \sigma_+ \sigma_- - i \hbar \gamma \sigma_+ \sigma_- + \hbar g (\sigma_+ + \sigma_-)$, $\sigma_- = \sigma \wedge \sigma_+ = \sigma^\dagger$

$$\Rightarrow K = \underbrace{\hbar(\omega - i\gamma) \sigma^\dagger \sigma}_{K_0} + \underbrace{\hbar g (\sigma^\dagger + \sigma)}_{K'}$$

Ec. maestra de Lindblad: $\mathcal{L}\rho = -\frac{i}{\hbar} (K\rho - \rho K^\dagger) + 2\gamma \sigma \rho \sigma^\dagger$

$$\begin{aligned} \Rightarrow \mathcal{L}\rho &= -\frac{i}{\hbar} [(K_0 + K')\rho - \rho(K_0^\dagger + K'^\dagger)] + 2\gamma \sigma \rho \sigma^\dagger \\ &= -\frac{i}{\hbar} (K_0 \rho + K' \rho - \rho K_0^\dagger - \rho K'^\dagger) + 2\gamma \sigma \rho \sigma^\dagger \\ &= \underbrace{-\frac{i}{\hbar} (K_0 \rho - \rho K_0^\dagger)}_{\mathcal{L}_0 \rho} + 2\gamma \sigma \rho \sigma^\dagger + \underbrace{-\frac{i}{\hbar} (K' \rho - \rho K'^\dagger)}_{V\rho} \end{aligned}$$

Formas matriciales:

$$\mathcal{L}_0 = \begin{pmatrix} 0 & 0 & 0 & 2\gamma \\ 0 & -\gamma + i\omega & 0 & 0 \\ 0 & 0 & -\gamma - i\omega & 0 \\ 0 & 0 & 0 & -2\gamma \end{pmatrix}$$

$$\begin{aligned} V &= -\frac{i}{\hbar} (K' \otimes \mathbb{1} - \mathbb{1} \otimes K'^\dagger) = -ig [(\sigma^\dagger + \sigma) \otimes \mathbb{1} - \mathbb{1} \otimes (\sigma + \sigma^\dagger)] , \quad \sigma^\dagger + \sigma = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= -ig \left[\begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \right] = -ig \begin{pmatrix} 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \end{pmatrix} \end{aligned}$$

j	λ_j	$\hat{\rho}_{\lambda_j}^{(0)}$	$\check{\rho}_{\lambda_j}^{(0)}$
1	0	$ g\rangle\langle g $	$\mathbb{1}$
2	-2γ	σ_z	$ e\rangle\langle e $
3	$-\gamma + i\omega$	$ g\rangle\langle e $	$ g\rangle\langle e $
4	$-\gamma - i\omega$	$ e\rangle\langle g $	$ e\rangle\langle g $

Queremos resolver $\mathcal{L} \hat{\rho}_{\lambda_j} = \lambda_j \hat{\rho}_{\lambda_j}$ y $\mathcal{L}^\dagger \check{\rho}_{\lambda_j} = \lambda_j^* \check{\rho}_{\lambda_j}$, donde $\mathcal{L} = \mathcal{L}_0 + \delta V$

$$\lambda_j = \lambda_j^{(0)} + \delta \lambda_j^{(1)} + \delta^2 \lambda_j^{(2)} + \dots, \quad \hat{\rho}_{\lambda_j} = \hat{\rho}_{\lambda_j}^{(0)} + \delta \hat{\rho}_{\lambda_j}^{(1)} + \delta^2 \hat{\rho}_{\lambda_j}^{(2)} + \dots, \quad \check{\rho}_{\lambda_j} = \check{\rho}_{\lambda_j}^{(0)} + \delta \check{\rho}_{\lambda_j}^{(1)} + \delta^2 \check{\rho}_{\lambda_j}^{(2)} + \dots$$

Antes de obtener las correcciones de 1^{er} TLS veamos como actúa la perturbación V sobre los eigenvectores de $\mathbb{I}_0$.

$$V\rho = -\frac{i}{\hbar} (K'\rho - \rho K') = -ig [(\sigma^+ + \sigma)\rho - \rho(\sigma^+ + \sigma)]$$

$$V\hat{\rho}_{\lambda_1}^{(0)} = -ig [(\sigma^+ + \sigma)|\rho_X g\rangle - |\rho_X g\rangle(\sigma^+ + \sigma)] = -ig (|\rho_X g\rangle - |\rho_X e\rangle)$$

$$\begin{aligned} V\hat{\rho}_{\lambda_2}^{(0)} &= -ig [(\sigma^+ + \sigma)(|\rho_X e\rangle - |\rho_X g\rangle) - (|\rho_X e\rangle - |\rho_X g\rangle)(\sigma^+ + \sigma)] \\ &= -ig (-|\rho_X g\rangle + |\rho_X e\rangle - |\rho_X g\rangle + |\rho_X e\rangle) = -2ig (|\rho_X e\rangle - |\rho_X g\rangle) \end{aligned}$$

$$V\hat{\rho}_{\lambda_3}^{(0)} = -ig [(\sigma^+ + \sigma)|\rho_X e\rangle - |\rho_X e\rangle(\sigma^+ + \sigma)] = -ig (|\rho_X e\rangle - |\rho_X g\rangle)$$

$$V\hat{\rho}_{\lambda_4}^{(0)} = -ig [(\sigma^+ + \sigma)|\rho_X g\rangle - |\rho_X g\rangle(\sigma^+ + \sigma)] = -ig (|\rho_X g\rangle - |\rho_X e\rangle)$$

$$\Rightarrow V\hat{\rho}_{\lambda_1}^{(0)} = -ig (|\rho_X g\rangle - |\rho_X e\rangle), \quad V\hat{\rho}_{\lambda_2}^{(0)} = -2ig (|\rho_X e\rangle - |\rho_X g\rangle),$$

$$V\hat{\rho}_{\lambda_3}^{(0)} = -ig (|\rho_X e\rangle - |\rho_X g\rangle), \quad V\hat{\rho}_{\lambda_4}^{(0)} = -ig (|\rho_X g\rangle - |\rho_X e\rangle)$$

Corrección de eigenvalores derechos a 1^{er} orden:

$$\lambda_j^{(1)} = \text{Tr} [\check{\rho}_{\lambda_j}^{(0)\dagger} V \hat{\rho}_{\lambda_j}^{(0)}]$$

$$\bullet \lambda_1^{(1)} = \text{Tr} [\check{\rho}_{\lambda_1}^{(0)\dagger} V \hat{\rho}_{\lambda_1}^{(0)}] = \text{Tr} \{ \mathbb{1} [-ig (|\rho_X g\rangle - |\rho_X e\rangle)] \} = -ig \text{Tr} (|\rho_X g\rangle - |\rho_X e\rangle) = 0$$

$$\bullet \lambda_2^{(1)} = \text{Tr} [\check{\rho}_{\lambda_2}^{(0)\dagger} V \hat{\rho}_{\lambda_2}^{(0)}] = -2ig \text{Tr} [|\rho_X e\rangle (|\rho_X e\rangle - |\rho_X g\rangle)] = -2ig \text{Tr} (-|\rho_X g\rangle) = 0$$

$$\bullet \lambda_3^{(1)} = \text{Tr} [\check{\rho}_{\lambda_3}^{(0)\dagger} V \hat{\rho}_{\lambda_3}^{(0)}] = -ig \text{Tr} [|\rho_X g\rangle (|\rho_X e\rangle - |\rho_X g\rangle)] = -ig \text{Tr} (-|\rho_X g\rangle) = 0$$

$$\bullet \lambda_4^{(1)} = \text{Tr} [\check{\rho}_{\lambda_4}^{(0)\dagger} V \hat{\rho}_{\lambda_4}^{(0)}] = -ig \text{Tr} [|\rho_X e\rangle (|\rho_X g\rangle - |\rho_X e\rangle)] = -ig \text{Tr} (-|\rho_X e\rangle) = 0$$

$$\Rightarrow \lambda_1^{(1)} = \lambda_2^{(1)} = \lambda_3^{(1)} = \lambda_4^{(1)} = 0$$

Corrección de eigenvectores derechos a 1^{er} orden:

$$\begin{aligned} \bullet \hat{\rho}_{\lambda_1}^{(1)} &= \sum_{i \neq 1} \frac{\text{Tr} (\check{\rho}_{\lambda_1}^{(0)\dagger} V \hat{\rho}_{\lambda_i}^{(0)})}{\lambda_1^{(0)} - \lambda_i^{(0)}} \hat{\rho}_{\lambda_i}^{(0)} \\ &= -ig \left\{ \frac{\text{Tr} [|\rho_X e\rangle (|\rho_X g\rangle - |\rho_X e\rangle)]}{0 - (-2\gamma)} \sigma_z + \frac{\text{Tr} [|\rho_X g\rangle (|\rho_X g\rangle - |\rho_X e\rangle)]}{0 - (-\gamma + i\omega)} |\rho_X e\rangle \right. \\ &\quad \left. + \frac{\text{Tr} [|\rho_X e\rangle (|\rho_X g\rangle - |\rho_X e\rangle)]}{0 - (-\gamma - i\omega)} |\rho_X g\rangle \right\} \end{aligned}$$

$$= -ig \left(\frac{|g\rangle\langle e|}{-\gamma + i\omega} - \frac{|e\rangle\langle g|}{-\gamma - i\omega} \right) = g \left(\frac{|g\rangle\langle e|}{-i\gamma - \omega} - \frac{|e\rangle\langle g|}{-i\gamma + \omega} \right) = g \left(\frac{|e\rangle\langle g|}{i\gamma - \omega} - \frac{|g\rangle\langle e|}{i\gamma + \omega} \right)$$

$$\begin{aligned} \hat{\rho}_{\lambda_2}^{(1)} &= \sum_{i \neq 2} \frac{\text{Tr}(\hat{\rho}_{\lambda_i}^{(0)\dagger} \hat{V} \hat{\rho}_{\lambda_2}^{(0)})}{\lambda_2^{(0)} - \lambda_i^{(0)}} \hat{\rho}_{\lambda_i}^{(0)} \\ &= -2ig \left\{ \frac{\text{Tr}[1(|g\rangle\langle e| - |e\rangle\langle g|)]}{-2\gamma - (0)} |g\rangle\langle g| + \frac{\text{Tr}[|e\rangle\langle g|(|g\rangle\langle e| - |e\rangle\langle g|)]}{-2\gamma - (-\gamma + i\omega)} |g\rangle\langle e| \right. \\ &\quad \left. + \frac{\text{Tr}[|g\rangle\langle e|(|g\rangle\langle e| - |e\rangle\langle g|)]}{-2\gamma - (-\gamma - i\omega)} |e\rangle\langle g| \right\} \\ &= -2ig \left(\frac{|g\rangle\langle e|}{-\gamma - i\omega} - \frac{|e\rangle\langle g|}{-\gamma + i\omega} \right) = 2g \left(\frac{|g\rangle\langle e|}{-i\gamma + \omega} - \frac{|e\rangle\langle g|}{-i\gamma - \omega} \right) = 2g \left(\frac{|e\rangle\langle g|}{i\gamma + \omega} - \frac{|g\rangle\langle e|}{i\gamma - \omega} \right) \end{aligned}$$

$$\begin{aligned} \hat{\rho}_{\lambda_3}^{(1)} &= \sum_{i \neq 3} \frac{\text{Tr}(\hat{\rho}_{\lambda_i}^{(0)\dagger} \hat{V} \hat{\rho}_{\lambda_3}^{(0)})}{\lambda_3^{(0)} - \lambda_i^{(0)}} \hat{\rho}_{\lambda_i}^{(0)} \\ &= -ig \left\{ \frac{\text{Tr}[1(|e\rangle\langle e| - |g\rangle\langle g|)]}{-\gamma + i\omega - (0)} |g\rangle\langle g| + \frac{\text{Tr}[|e\rangle\langle e|(|e\rangle\langle e| - |g\rangle\langle g|)]}{-\gamma + i\omega - (-2\gamma)} \sigma_z \right. \\ &\quad \left. + \frac{\text{Tr}[|g\rangle\langle e|(|e\rangle\langle e| - |g\rangle\langle g|)]}{-\gamma + i\omega - (-\gamma - i\omega)} |e\rangle\langle g| \right\} \\ &= -ig \frac{|e\rangle\langle e| - |g\rangle\langle g|}{\gamma + i\omega} = \frac{g}{i\gamma - \omega} (|e\rangle\langle e| - |g\rangle\langle g|) \end{aligned}$$

$$\begin{aligned} \hat{\rho}_{\lambda_4}^{(1)} &= \sum_{i \neq 4} \frac{\text{Tr}(\hat{\rho}_{\lambda_i}^{(0)\dagger} \hat{V} \hat{\rho}_{\lambda_4}^{(0)})}{\lambda_4^{(0)} - \lambda_i^{(0)}} \hat{\rho}_{\lambda_i}^{(0)} \\ &= -ig \left\{ \frac{\text{Tr}[1(|g\rangle\langle g| - |e\rangle\langle e|)]}{-\gamma - i\omega - (0)} |g\rangle\langle g| + \frac{\text{Tr}[|e\rangle\langle e|(|g\rangle\langle g| - |e\rangle\langle e|)]}{-\gamma - i\omega - (-2\gamma)} \sigma_z \right. \\ &\quad \left. + \frac{\text{Tr}[|e\rangle\langle g|(|g\rangle\langle g| - |e\rangle\langle e|)]}{-\gamma - i\omega - (-\gamma + i\omega)} |g\rangle\langle e| \right\} \\ &= -ig \frac{-1}{\gamma - i\omega} \sigma_z = \frac{g}{i\gamma + \omega} (|g\rangle\langle g| - |e\rangle\langle e|) \end{aligned}$$

$$\Rightarrow \begin{aligned} \hat{\rho}_{\lambda_1}^{(1)} &= g \left(\frac{|e\rangle\langle g|}{i\gamma - \omega} - \frac{|g\rangle\langle e|}{i\gamma + \omega} \right), & \hat{\rho}_{\lambda_2}^{(1)} &= 2g \left(\frac{|e\rangle\langle g|}{i\gamma + \omega} - \frac{|g\rangle\langle e|}{i\gamma - \omega} \right), \\ \hat{\rho}_{\lambda_3}^{(1)} &= \frac{g}{i\gamma - \omega} (|e\rangle\langle e| - |g\rangle\langle g|), & \hat{\rho}_{\lambda_4}^{(1)} &= \frac{g}{i\gamma + \omega} (|g\rangle\langle g| - |e\rangle\langle e|) \end{aligned}$$

Corrección de eigenvalores derechos a 2º orden:

$$\lambda_j^{(2)} = \text{Tr}(\check{\rho}_{\lambda_j}^{(0)\dagger} V \hat{\rho}_{\lambda_j}^{(0)})$$

$$\begin{aligned} \bullet \lambda_1^{(2)} &= \text{Tr}(\check{\rho}_{\lambda_1}^{(0)\dagger} V \hat{\rho}_{\lambda_1}^{(0)}) = \text{Tr}\left\{ \mathbb{1} V \left[g \left(\frac{|exg\rangle}{i\tau - \omega} - \frac{|gxel\rangle}{i\tau + \omega} \right) \right] \right\} = \frac{g}{i\tau - \omega} \text{Tr}(V|exg\rangle) - \frac{g}{i\tau + \omega} \text{Tr}(V|gxel\rangle) \\ &= \frac{g}{i\tau - \omega} \text{Tr}[-ig(|gxg\rangle - |xel\rangle)] - \frac{g}{i\tau + \omega} \text{Tr}[-ig(|xel\rangle - |gxg\rangle)] = 0 \end{aligned}$$

$$\begin{aligned} \bullet \lambda_2^{(2)} &= \text{Tr}(\check{\rho}_{\lambda_2}^{(0)\dagger} V \hat{\rho}_{\lambda_2}^{(0)}) = \text{Tr}\left\{ |xel\rangle V \left[zg \left(\frac{|exg\rangle}{i\tau - \omega} - \frac{|gxel\rangle}{i\tau + \omega} \right) \right] \right\} \\ &= \frac{zg}{i\tau - \omega} \text{Tr}[|xel\rangle V(|exg\rangle)] - \frac{zg}{i\tau + \omega} \text{Tr}[|xel\rangle V(|gxel\rangle)] \\ &= \frac{zg}{i\tau - \omega} \text{Tr}[-ig|xel\rangle(|gxg\rangle - |xel\rangle)] - \frac{zg}{i\tau + \omega} \text{Tr}[-ig|xel\rangle(|xel\rangle - |gxg\rangle)] \\ &= \frac{zig^2}{i\tau - \omega} + \frac{zig^2}{i\tau + \omega} = zg^2 \left(\frac{1}{\tau + i\omega} + \frac{1}{\tau - i\omega} \right) = zg^2 \left(\frac{2\tau}{\tau^2 + \omega^2} \right) = \frac{4g^2\tau}{\tau^2 + \omega^2} \\ &\quad \hookrightarrow \frac{1}{a+ib} + \frac{1}{a-ib} = \frac{2a}{a^2+b^2} \end{aligned}$$

$$\begin{aligned} \bullet \lambda_3^{(2)} &= \text{Tr}(\check{\rho}_{\lambda_3}^{(0)\dagger} V \hat{\rho}_{\lambda_3}^{(0)}) = \text{Tr}\left\{ |exg\rangle V \left[\frac{g}{i\tau - \omega} (|xel\rangle - |gxg\rangle) \right] \right\} \\ &= \frac{g}{i\tau - \omega} \left[\text{Tr}(|exg\rangle V|xel\rangle) - \text{Tr}(|exg\rangle V|gxg\rangle) \right] \\ &\quad -ig[(\sigma^+ + \sigma)|xel\rangle - |xel\rangle(\sigma^+ + \sigma)] = -ig(|gxel\rangle - |exg\rangle) \\ &= \frac{-ig^2}{i\tau - \omega} \left\{ \text{Tr}[|exg\rangle(|gxel\rangle - |exg\rangle)] - \text{Tr}[|exg\rangle(|exg\rangle - |gxel\rangle)] \right\} = \frac{-i2g^2}{i\tau - \omega} \\ &= \frac{2g^2}{-\tau - i\omega} = -\frac{2g^2}{\tau + i\omega} \end{aligned}$$

$$\begin{aligned} \bullet \lambda_4^{(2)} &= \text{Tr}(\check{\rho}_{\lambda_4}^{(0)\dagger} V \hat{\rho}_{\lambda_4}^{(0)}) = \text{Tr}\left\{ |gxel\rangle V \left[\frac{g}{i\tau + \omega} (|gxg\rangle - |xel\rangle) \right] \right\} \\ &= \frac{g}{i\tau + \omega} \left[\text{Tr}(|gxel\rangle V|gxg\rangle) - \text{Tr}(|gxel\rangle V|xel\rangle) \right] \\ &= \frac{-ig^2}{i\tau + \omega} \left\{ \text{Tr}[|gxel\rangle(|exg\rangle - |gxel\rangle)] - \text{Tr}[|gxel\rangle(|gxel\rangle - |exg\rangle)] \right\} = \frac{-i2g^2}{i\tau + \omega} \\ &= \frac{2g^2}{-\tau + i\omega} \end{aligned}$$

$$\Rightarrow \lambda_1^{(2)} = 0, \quad \lambda_2^{(2)} = \frac{4g^2\tau}{\tau^2 + \omega^2}, \quad \lambda_3^{(2)} = \frac{2g^2}{-\tau - i\omega}, \quad \lambda_4^{(2)} = \frac{2g^2}{-\tau + i\omega}$$

Corrección de eigenvalores izquierdos a 1º orden:

$$\tilde{\lambda}_j^{(1)} = \text{Tr}(\check{\rho}_{\lambda_j}^{(0)\dagger} V \hat{\rho}_{\lambda_j}^{(0)})^* = \lambda_j^{(1)*} \Rightarrow \tilde{\lambda}_1^{(1)} = \tilde{\lambda}_2^{(1)} = \tilde{\lambda}_3^{(1)} = \tilde{\lambda}_4^{(1)} = 0$$

Corrección de eigenvectores izquierdos a 1^{er} orden:

$$\check{\gamma}_{\lambda_j}^{(1)} = \sum_{i \neq j} \left[\frac{\text{Tr}(\check{\gamma}_{\lambda_j}^{(0)\dagger} V \hat{\gamma}_{\lambda_i}^{(0)})}{(\lambda_j^{(0)} - \lambda_i^{(0)})} \right]^* \check{\gamma}_{\lambda_i}^{(0)}$$

$$\bullet \check{\gamma}_{\lambda_1}^{(1)} = \sum_{i \neq 1} \left[\frac{\text{Tr}(\check{\gamma}_{\lambda_1}^{(0)\dagger} V \hat{\gamma}_{\lambda_i}^{(0)})}{(\lambda_1^{(0)} - \lambda_i^{(0)})} \right]^* \check{\gamma}_{\lambda_i}^{(0)} = 0$$

$$\bullet \check{\gamma}_{\lambda_2}^{(1)} = \sum_{i \neq 2} \left[\frac{\text{Tr}(\check{\gamma}_{\lambda_2}^{(0)\dagger} V \hat{\gamma}_{\lambda_i}^{(0)})}{(\lambda_2^{(0)} - \lambda_i^{(0)})} \right]^* \check{\gamma}_{\lambda_i}^{(0)}$$

$$= \left[\frac{-ig}{-2\gamma - (-\gamma + i\omega)} \right]^* |gxe\rangle + \left[\frac{ig}{-2\gamma - (-\gamma - i\omega)} \right]^* |exg\rangle$$

$$= \left(\frac{-ig}{-\gamma - i\omega} \right)^* |gxe\rangle + \left(\frac{ig}{-\gamma + i\omega} \right)^* |exg\rangle = \left(\frac{g}{-i\gamma + \omega} \right)^* |gxe\rangle + \left(\frac{g}{i\gamma + \omega} \right)^* |exg\rangle$$

$$= g \left(\frac{|gxe\rangle}{i\gamma + \omega} + \frac{|exg\rangle}{-i\gamma + \omega} \right) = g \left(\frac{|gxe\rangle}{i\gamma + \omega} - \frac{|exg\rangle}{i\gamma - \omega} \right)$$

$$\bullet \check{\gamma}_{\lambda_3}^{(1)} = \sum_{i \neq 3} \left[\frac{\text{Tr}(\check{\gamma}_{\lambda_3}^{(0)\dagger} V \hat{\gamma}_{\lambda_i}^{(0)})}{(\lambda_3^{(0)} - \lambda_i^{(0)})} \right]^* \check{\gamma}_{\lambda_i}^{(0)}$$

$$= \left(\frac{ig}{-\gamma + i\omega} \right)^* \mathbb{1} + \left(\frac{-2ig}{\gamma + i\omega} \right)^* |ex\rangle = \left(\frac{g}{i\gamma + \omega} \right)^* \mathbb{1} + \left(\frac{2g}{i\gamma - \omega} \right)^* |ex\rangle$$

$$= g \left(\frac{\mathbb{1}}{-i\gamma + \omega} + \frac{2|ex\rangle}{-i\gamma - \omega} \right)$$

$$\bullet \check{\gamma}_{\lambda_4}^{(1)} = \sum_{i \neq 4} \left[\frac{\text{Tr}(\check{\gamma}_{\lambda_4}^{(0)\dagger} V \hat{\gamma}_{\lambda_i}^{(0)})}{(\lambda_4^{(0)} - \lambda_i^{(0)})} \right]^* \check{\gamma}_{\lambda_i}^{(0)}$$

$$= \left(\frac{-ig}{-\gamma - i\omega} \right)^* \mathbb{1} + \left(\frac{2ig}{\gamma - i\omega} \right)^* |ex\rangle = \left(\frac{g}{-i\gamma + \omega} \right)^* \mathbb{1} + \left(\frac{2g}{-i\gamma - \omega} \right)^* |ex\rangle$$

$$= g \left(\frac{\mathbb{1}}{i\gamma + \omega} + \frac{2|ex\rangle}{i\gamma - \omega} \right)$$

$$\Rightarrow \check{\gamma}_{\lambda_1}^{(1)} = 0,$$

$$\check{\gamma}_{\lambda_2}^{(1)} = g \left(\frac{|gxe\rangle}{i\gamma + \omega} - \frac{|exg\rangle}{i\gamma - \omega} \right),$$

$$\check{\gamma}_{\lambda_3}^{(1)} = g \left(\frac{\mathbb{1}}{-i\gamma + \omega} + \frac{2|ex\rangle}{-i\gamma - \omega} \right), \quad \check{\gamma}_{\lambda_4}^{(1)} = g \left(\frac{\mathbb{1}}{i\gamma + \omega} + \frac{2|ex\rangle}{i\gamma - \omega} \right)$$

Corrección de eigenvalores izquierdos a 2^{do} orden:

$$\tilde{\lambda}_j^{(2)} = \left[\sum_{i \neq j} \frac{\text{Tr}(\check{\rho}_{\lambda_j}^{(0)\dagger} V \hat{\rho}_{\lambda_i}^{(0)}) \text{Tr}(\check{\rho}_{\lambda_i}^{(0)\dagger} V \hat{\rho}_{\lambda_j}^{(0)})}{\lambda_j^{(0)} - \lambda_i^{(0)}} \right]^* = \lambda_j^{(2)*}$$

$$\Rightarrow \tilde{\lambda}_1^{(2)} = 0, \quad \tilde{\lambda}_2^{(2)} = \frac{4g^2\gamma}{\gamma^2 + \omega^2}, \quad \tilde{\lambda}_3^{(2)} = \frac{2g^2}{-\gamma + i\omega}, \quad \tilde{\lambda}_4^{(2)} = \frac{2g^2}{-\gamma - i\omega}$$

Finalmente al 1^{er} orden no nulo los eigenvalores y eigenvectores de $\mathcal{L}$ son:

$$\lambda_1 = 0 \quad \hat{\rho}_{\lambda_1} = |\gamma \times g\rangle + \delta g \left(\frac{|\epsilon \times g\rangle}{i\gamma - \omega} - \frac{|\gamma \times \epsilon\rangle}{i\gamma + \omega} \right)$$

$$\lambda_2 = -2\gamma + \delta^2 \frac{4g^2\gamma}{\gamma^2 + \omega^2} \quad \hat{\rho}_{\lambda_2} = \sigma_z + 2\delta g \left(\frac{|\epsilon \times g\rangle}{i\gamma + \omega} - \frac{|\gamma \times \epsilon\rangle}{i\gamma - \omega} \right)$$

$$\lambda_3 = -\gamma + i\omega + \delta^2 \frac{2g^2}{-\gamma - i\omega} \quad \hat{\rho}_{\lambda_3} = |\gamma \times \epsilon\rangle + \frac{\delta g}{i\gamma - \omega} (|\epsilon \times \epsilon\rangle - |\gamma \times g\rangle)$$

$$\lambda_4 = -\gamma - i\omega + \delta^2 \frac{2g^2}{-\gamma + i\omega} \quad \hat{\rho}_{\lambda_4} = |\epsilon \times g\rangle + \frac{\delta g}{i\gamma + \omega} (|\gamma \times g\rangle - |\epsilon \times \epsilon\rangle)$$

$$\tilde{\lambda}_1 = 0 \quad \check{\rho}_{\lambda_1} = \mathbb{1}$$

$$\tilde{\lambda}_2 = -2\gamma + \delta^2 \frac{4g^2\gamma}{\gamma^2 + \omega^2} \quad \check{\rho}_{\lambda_2} = |\epsilon \times \epsilon\rangle + \delta g \left(\frac{|\gamma \times \epsilon\rangle}{i\gamma + \omega} - \frac{|\epsilon \times g\rangle}{i\gamma - \omega} \right)$$

$$\tilde{\lambda}_3 = -\gamma - i\omega + \delta^2 \frac{2g^2}{-\gamma + i\omega} \quad \check{\rho}_{\lambda_3} = |\gamma \times \epsilon\rangle + \delta g \left(\frac{\mathbb{1}}{-i\gamma + \omega} + \frac{2|\epsilon \times \epsilon\rangle}{-i\gamma - \omega} \right)$$

$$\tilde{\lambda}_4 = -\gamma + i\omega + \delta^2 \frac{2g^2}{-\gamma - i\omega} \quad \check{\rho}_{\lambda_4} = |\epsilon \times g\rangle + \delta g \left(\frac{\mathbb{1}}{i\gamma + \omega} + \frac{2|\epsilon \times \epsilon\rangle}{i\gamma - \omega} \right)$$

j	λ_j	$\hat{\rho}_{\lambda_j}$	$\check{\rho}_{\lambda_j}$
1	0	$ \gamma \times g\rangle + \delta g \left(\frac{ \epsilon \times g\rangle}{i\gamma - \omega} - \frac{ \gamma \times \epsilon\rangle}{i\gamma + \omega} \right)$	$\mathbb{1}$
2	$-2\gamma + \delta^2 \frac{4g^2\gamma}{\gamma^2 + \omega^2}$	$\sigma_z + 2\delta g \left(\frac{ \epsilon \times g\rangle}{i\gamma + \omega} - \frac{ \gamma \times \epsilon\rangle}{i\gamma - \omega} \right)$	$ \epsilon \times \epsilon\rangle + \delta g \left(\frac{ \gamma \times \epsilon\rangle}{i\gamma + \omega} - \frac{ \epsilon \times g\rangle}{i\gamma - \omega} \right)$
3	$-\gamma + i\omega + \delta^2 \frac{2g^2}{-\gamma - i\omega}$	$ \gamma \times \epsilon\rangle + \frac{\delta g}{i\gamma - \omega} (\epsilon \times \epsilon\rangle - \gamma \times g\rangle)$	$ \gamma \times \epsilon\rangle + \delta g \left(\frac{\mathbb{1}}{-i\gamma + \omega} + \frac{2 \epsilon \times \epsilon\rangle}{-i\gamma - \omega} \right)$
4	$-\gamma - i\omega + \delta^2 \frac{2g^2}{-\gamma + i\omega}$	$ \epsilon \times g\rangle + \frac{\delta g}{i\gamma + \omega} (\gamma \times g\rangle - \epsilon \times \epsilon\rangle)$	$ \epsilon \times g\rangle + \delta g \left(\frac{\mathbb{1}}{i\gamma + \omega} + \frac{2 \epsilon \times \epsilon\rangle}{i\gamma - \omega} \right)$

Productos internos

Hemos impuesto que $\text{Tr}(\check{\rho}_{\lambda_j}^{(0)+} \hat{\rho}_{\lambda_i}) = 1$

$$\Rightarrow \text{Tr}(\check{\rho}_{\lambda_1}^{(0)+} \hat{\rho}_{\lambda_1}) = \text{Tr} \left\{ \mathbb{1} \left[|gXg| + \delta g \left(\frac{|cXg|}{i\tau - \omega} - \frac{|gXe|}{i\tau + \omega} \right) \right] \right\} = 1$$

$$\text{Tr}(\check{\rho}_{\lambda_2}^{(0)+} \hat{\rho}_{\lambda_2}) = \text{Tr} \left\{ |Xe| \left[\sigma_z + 2\delta g \left(\frac{|cXg|}{i\tau + \omega} - \frac{|gXe|}{i\tau - \omega} \right) \right] \right\} = 1$$

$$\text{Tr}(\check{\rho}_{\lambda_3}^{(0)+} \hat{\rho}_{\lambda_3}) = \text{Tr} \left\{ |eXg| \left[|gXe| + \frac{\delta g}{i\tau - \omega} (|Xe| - |gXg|) \right] \right\} = 1$$

$$\text{Tr}(\check{\rho}_{\lambda_4}^{(0)+} \hat{\rho}_{\lambda_4}) = \text{Tr} \left\{ |gXe| \left[|eXg| + \frac{\delta g}{i\tau + \omega} (|gXg| - |Xe|) \right] \right\} = 1$$

Veamos los productos internos de Hilbert-Schmidt:

$$\text{Tr}(\check{\rho}_{\lambda_j}^+ \hat{\rho}_{\lambda_i}) = \delta_{\lambda_j, \lambda_i}$$

$$\text{Tr}(\check{\rho}_{\lambda_1}^+ \hat{\rho}_{\lambda_1}) = \text{Tr} \left\{ \mathbb{1} \left[|gXg| + \delta g \left(\frac{|cXg|}{i\tau - \omega} - \frac{|gXe|}{i\tau + \omega} \right) \right] \right\} = 1$$

$$\begin{aligned} \text{Tr}(\check{\rho}_{\lambda_2}^+ \hat{\rho}_{\lambda_2}) &= \text{Tr} \left\{ \left[|Xe| + \delta g \left(\frac{|gXe|}{i\tau + \omega} - \frac{|eXg|}{i\tau - \omega} \right) \right]^\dagger \left[\sigma_z + 2\delta g \left(\frac{|cXg|}{i\tau + \omega} - \frac{|gXe|}{i\tau - \omega} \right) \right] \right\} \\ &= \text{Tr} \left\{ \left[|Xe| + \delta g \left(\frac{|cXg|}{-i\tau + \omega} - \frac{|gXe|}{-i\tau - \omega} \right) \right] \left[\sigma_z + 2\delta g \left(\frac{|cXg|}{i\tau + \omega} - \frac{|gXe|}{i\tau - \omega} \right) \right] \right\} \\ &= 1 - \frac{2\delta^2 g^2}{(-i\tau + \omega)(i\tau - \omega)} - \frac{2\delta^2 g^2}{(-i\tau - \omega)(i\tau + \omega)} \\ &= 1 - \frac{2\delta^2 g^2}{(\tau + i\omega)^2} + \frac{2\delta^2 g^2}{(i\tau + \omega)^2} \end{aligned}$$

$$\begin{aligned} \text{Tr}(\check{\rho}_{\lambda_3}^+ \hat{\rho}_{\lambda_3}) &= \text{Tr} \left\{ \left[|eXg| + \delta g \left(\frac{\mathbb{1}}{i\tau + \omega} + \frac{2|Xe|}{i\tau - \omega} \right) \right] \left[|gXe| + \frac{\delta g}{i\tau - \omega} (|Xe| - |gXg|) \right] \right\} \\ &= 1 + \frac{2\delta^2 g^2}{(i\tau - \omega)^2} \end{aligned}$$

$$\begin{aligned} \text{Tr}(\check{\rho}_{\lambda_4}^+ \hat{\rho}_{\lambda_4}) &= \text{Tr} \left\{ \left[|gXe| + \delta g \left(\frac{\mathbb{1}}{-i\tau + \omega} + \frac{2|Xe|}{-i\tau - \omega} \right) \right] \left[|eXg| + \frac{\delta g}{i\tau + \omega} (|gXg| - |Xe|) \right] \right\} \\ &= 1 + \frac{2\delta^2 g^2}{(i\tau + \omega)^2} \end{aligned}$$

Los términos con δ^2 son errores de truncamiento por lo que no hace falta renormalizar.

Estado estacionario

El estado estacionario de este sistema es $\hat{y}_{ss} = 19 \times 91 + \varepsilon \left(\frac{10 \times 91}{i\tau - \omega} - \frac{19 \times e1}{2\tau + \omega} \right)$

Pues es el estado físico con eigenvalor 0, Queremos ahora calcularlo analíticamente y compararlo con el resultado perturbativo.

Sabemos que $\mathcal{L}\hat{y}_{ss} = 0$ para el estado estacionario y la forma matricial de $\mathcal{L}$ es:

$$\mathcal{L} = \mathcal{L}_0 + \varepsilon V = \begin{pmatrix} 0 & 0 & 0 & 2\tau \\ 0 & -\tau + i\omega & 0 & 0 \\ 0 & 0 & -\tau - i\omega & 0 \\ 0 & 0 & 0 & -2\tau \end{pmatrix} - \varepsilon i \begin{pmatrix} 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \varepsilon i & -\varepsilon i & 2\tau \\ \varepsilon i & -\tau + i\omega & 0 & -\varepsilon i \\ -\varepsilon i & 0 & -\tau - i\omega & \varepsilon i \\ 0 & -\varepsilon i & \varepsilon i & -2\tau \end{pmatrix}$$

$$\hat{y}_{ss} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & \varepsilon i & -\varepsilon i & 2\tau \\ \varepsilon i & -\tau + i\omega & 0 & -\varepsilon i \\ -\varepsilon i & 0 & -\tau - i\omega & \varepsilon i \\ 0 & -\varepsilon i & \varepsilon i & -2\tau \end{pmatrix} \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = \vec{0}$$

$$\Rightarrow \begin{cases} b\varepsilon i - c\varepsilon i + d2\tau = 0 & ① \\ a\varepsilon i + b(-\tau + i\omega) - d\varepsilon i = 0 & ② \\ -a\varepsilon i - c(\tau + i\omega) + d\varepsilon i = 0 & ③ \\ -b\varepsilon i + c\varepsilon i - d2\tau = 0 & ④ \end{cases}$$

$$① = ④$$

$$②: b(-\tau + i\omega) = -a\varepsilon i + d\varepsilon i \Rightarrow b = \frac{-a\varepsilon i + d\varepsilon i}{-\tau + i\omega}$$

$$③: -a\varepsilon i + d\varepsilon i = c(\tau + i\omega)$$

$$② + ③: b(-\tau + i\omega) = c(\tau + i\omega) \Rightarrow b = c \frac{\tau + i\omega}{-\tau + i\omega}$$

$$\frac{\tau - i\omega + \tau + i\omega}{\tau - i\omega}$$

↑

$$①: d2\tau = -b\varepsilon i + c\varepsilon i = -c \frac{\tau + i\omega}{-\tau + i\omega} \varepsilon i + c\varepsilon i = c\varepsilon i \left[1 + \frac{\tau + i\omega}{\tau - i\omega} \right]$$

$$= c\varepsilon i \frac{2\tau}{\tau - i\omega} \Rightarrow d = \frac{c\varepsilon i}{\tau - i\omega}$$

$$\textcircled{3}: -a \varepsilon i - c(\gamma + i\omega) + \frac{c \varepsilon i}{\gamma - i\omega} \varepsilon i = 0 \Rightarrow -a \varepsilon i - c(\gamma + i\omega) - \frac{c \varepsilon^2}{\gamma - i\omega} \Rightarrow -a \varepsilon i = c \left(\gamma + i\omega + \frac{\varepsilon^2}{\gamma - i\omega} \right)$$

$$\Rightarrow a = \frac{i}{\varepsilon} c \left(\gamma + i\omega + \frac{\varepsilon^2}{\gamma - i\omega} \right) = \frac{i}{\varepsilon} c \left[\frac{(\gamma + i\omega)(\gamma - i\omega) + \varepsilon^2}{\gamma - i\omega} \right] \Rightarrow a = \frac{ic(\varepsilon^2 + \gamma^2 + \omega^2)}{\varepsilon(\gamma - i\omega)}$$

$$\text{Sea } c=1: \hat{\mathcal{S}}_{ss} = \begin{pmatrix} i \frac{(\varepsilon^2 + \gamma^2 + \omega^2)}{\varepsilon(\gamma - i\omega)} & \frac{\gamma + i\omega}{-\gamma + i\omega} \\ 1 & \frac{\varepsilon i}{\gamma - i\omega} \end{pmatrix}$$

Multipliquemos por $\frac{-i \varepsilon (\gamma - i\omega)}{\gamma^2 + \omega^2}$:

$$a = i \frac{(\varepsilon^2 + \gamma^2 + \omega^2)}{\varepsilon(\gamma - i\omega)} \cdot \frac{-i \varepsilon (\gamma - i\omega)}{\gamma^2 + \omega^2} = 1 + \frac{\varepsilon^2}{\gamma^2 + \omega^2}$$

$$d = \frac{\varepsilon i}{\gamma - i\omega} \cdot \frac{-i \varepsilon (\gamma - i\omega)}{\gamma^2 + \omega^2} = \frac{\varepsilon^2}{\gamma^2 + \omega^2}$$

$$\hat{\mathcal{S}}_{ss} = \begin{pmatrix} \frac{\gamma^2 + \gamma^2 + \omega^2}{\gamma^2 + \omega^2} & \frac{i\gamma(\gamma + i\omega)}{\gamma^2 + \omega^2} \\ \frac{-i\gamma(\gamma - i\omega)}{\gamma^2 + \omega^2} & \frac{\varepsilon^2 \gamma^2}{\gamma^2 + \omega^2} \end{pmatrix} = \begin{pmatrix} 1 + \frac{\varepsilon^2 \gamma^2}{\gamma^2 + \omega^2} & \frac{i\gamma}{\gamma - i\omega} \\ -\frac{i\gamma}{\gamma + i\omega} & \frac{\varepsilon^2 \gamma^2}{\gamma^2 + \omega^2} \end{pmatrix}$$

$(\gamma + i\omega)(\gamma - i\omega)$

$$= \begin{pmatrix} 1 + \frac{\varepsilon^2 \gamma^2}{\gamma^2 + \omega^2} & -\frac{\gamma}{i\gamma + \omega} \\ \frac{\gamma}{i\gamma - \omega} & \frac{\varepsilon^2 \gamma^2}{\gamma^2 + \omega^2} \end{pmatrix}$$

A primer orden en ε :

$$\hat{\mathcal{S}}_{ss} = \begin{pmatrix} 1 & -\frac{\gamma}{i\gamma + \omega} \\ \frac{\gamma}{i\gamma - \omega} & 0 \end{pmatrix}$$

comparando con la sol. perturbativa vemos que son idénticas!

$$\hat{\mathcal{S}}_{\lambda_1} = |\mathbf{e} \times \mathbf{g}| + \varepsilon \mathbf{g} \left(\frac{|\mathbf{e} \times \mathbf{g}|}{i\gamma - \omega} - \frac{|\mathbf{g} \times \mathbf{e}|}{i\gamma + \omega} \right) = \begin{pmatrix} 1 & -\frac{\gamma}{i\gamma + \omega} \\ \frac{\gamma}{i\gamma - \omega} & 0 \end{pmatrix}$$

Dinámicas

Sabemos que la evolución temporal de una matriz de densidad es:

$$\rho(t) = \sum_n \text{Tr}(\hat{\rho}_n^\dagger \rho(0)) e^{\lambda_n t} \hat{\rho}_n$$

Calculamos la evolución temporal de $\rho(0) = |\text{ex}\rangle\langle\text{ex}|$:

$$\begin{aligned} \rho_{ee}(t) &= |\rho_{Xg}| + \delta \varepsilon \left(\frac{|\text{exg}|}{i\gamma - \omega} - \frac{|\rho_{Xe}|}{i\gamma + \omega} \right) \\ &+ e^{(-2\gamma + \varepsilon^2 \frac{4\gamma}{\gamma^2 + \omega^2})t} \left[|\text{ex}\rangle\langle\text{ex}| - |\rho_{Xg}| + 2\varepsilon \left(\frac{|\text{exg}|}{i\gamma + \omega} - \frac{|\rho_{Xe}|}{i\gamma - \omega} \right) \right] \\ &+ \varepsilon \left(\frac{1}{i\gamma + \omega} + \frac{2}{i\gamma - \omega} \right) e^{(-\gamma + i\omega + \varepsilon^2 \frac{2}{\gamma - i\omega})t} \left[|\rho_{Xe}| + \frac{\varepsilon}{i\gamma - \omega} (|\text{ex}\rangle\langle\text{ex}| - |\rho_{Xg}|) \right] \\ &+ \varepsilon \left(\frac{1}{-i\gamma + \omega} + \frac{2}{-i\gamma - \omega} \right) e^{(-\gamma - i\omega - \varepsilon^2 \frac{2}{\gamma + i\omega})t} \left[|\rho_{Xg}| + \frac{\varepsilon}{i\gamma + \omega} (|\rho_{Xe}| - |\text{ex}\rangle\langle\text{ex}|) \right] \end{aligned}$$

n	λ_n	$\hat{\rho}_n$	$\hat{I}_n$
1	0	$ \rho_{Xg} + \delta \varepsilon \left(\frac{ \text{exg} }{i\gamma - \omega} - \frac{ \rho_{Xe} }{i\gamma + \omega} \right)$	1
2	$-2\gamma + \varepsilon^2 \frac{4\gamma}{\gamma^2 + \omega^2}$	$ \text{ex}\rangle\langle\text{ex} - \rho_{Xg} + 2\varepsilon \left(\frac{ \text{exg} }{i\gamma + \omega} - \frac{ \rho_{Xe} }{i\gamma - \omega} \right)$	$ \text{ex}\rangle\langle\text{ex} + \delta \varepsilon \left(\frac{ \rho_{Xe} }{i\gamma + \omega} - \frac{ \rho_{Xg} }{i\gamma - \omega} \right)$
3	$-\gamma + i\omega + \varepsilon^2 \frac{2}{\gamma - i\omega}$	$ \rho_{Xe} + \frac{\varepsilon}{i\gamma - \omega} (\text{ex}\rangle\langle\text{ex} - \rho_{Xg})$	$ \rho_{Xe} + \delta \varepsilon \left(\frac{1}{-i\gamma + \omega} + \frac{2 \rho_{Xe} }{-i\gamma - \omega} \right)$
4	$-\gamma - i\omega - \varepsilon^2 \frac{2}{\gamma + i\omega}$	$ \rho_{Xg} + \frac{\varepsilon}{i\gamma + \omega} (\rho_{Xe} - \text{ex}\rangle\langle\text{ex})$	$ \rho_{Xg} + \delta \varepsilon \left(\frac{1}{i\gamma + \omega} + \frac{2 \rho_{Xg} }{i\gamma - \omega} \right)$

Ahora calculamos el valor esperado del esta excitado como función del tiempo

$$\begin{aligned} P_e = \langle |e\rangle \rangle &= \text{Tr}(\rho_{ee}(t) |\text{ex}\rangle\langle\text{ex}|) \\ &= e^{(-2\gamma + \varepsilon^2 \frac{4\gamma}{\gamma^2 + \omega^2})t} + \frac{\varepsilon}{i\gamma - \omega} \varepsilon \left(\frac{1}{i\gamma + \omega} + \frac{2}{i\gamma - \omega} \right) e^{(-\gamma + i\omega + \varepsilon^2 \frac{2}{\gamma - i\omega})t} \\ &- \frac{\varepsilon}{i\gamma + \omega} \varepsilon \left(\frac{1}{-i\gamma + \omega} + \frac{2}{-i\gamma - \omega} \right) e^{(-\gamma - i\omega - \varepsilon^2 \frac{2}{\gamma + i\omega})t} \end{aligned}$$

Ignorando términos de $O(\varepsilon^2)$ tenemos que $P_e = e^{-2\gamma t (1 - \frac{2\varepsilon^2}{\gamma^2 + \omega^2})}$

Similarmente para P_g obtenemos $P_g = 1 - e^{-2\gamma t (1 - \frac{2\varepsilon^2}{\gamma^2 + \omega^2})}$

Para las coherencias a partir de la matriz de densidad $\rho(0) = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$ tenemos:

$$P_{ge} = \frac{1}{2} e^{-\frac{\gamma^2 + \omega^2 + 2\varepsilon^2}{\gamma - i\omega} t} + i\varepsilon \left[\frac{1}{\gamma - i\omega} e^{-2\gamma t (1 - \frac{2\varepsilon^2}{\gamma^2 + \omega^2})} + \frac{1}{\gamma + i\omega} - \frac{2\gamma e^{-\frac{\gamma^2 + \omega^2 + 2\varepsilon^2}{\gamma - i\omega} t}}{\gamma^2 + \omega^2} \right]$$

Estado estacionario

El estado estacionario de este sistema es $\hat{\rho}_{ss} = |g\rangle\langle g| + \varepsilon \left(\frac{|e\rangle\langle g|}{i\gamma - \omega} - \frac{|g\rangle\langle e|}{i\gamma + \omega} \right)$

Pues es el estado físico con eigenvalor 0. Queremos ahora calcularlo analíticamente y compararlo con el resultado perturbativo.

Sabemos que $\mathcal{L}\hat{\rho}_{ss} = 0$ para el estado estacionario y la forma matricial de $\mathcal{L}$ es:

$$\mathcal{L} = \mathcal{L}_0 + \varepsilon V = \begin{pmatrix} 0 & 0 & 0 & 2\gamma \\ 0 & -\gamma + i\omega & 0 & 0 \\ 0 & 0 & -\gamma - i\omega & 0 \\ 0 & 0 & 0 & -2\gamma \end{pmatrix} - \varepsilon i \begin{pmatrix} 0 & -1 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & \varepsilon i & -\varepsilon i & 2\gamma \\ \varepsilon i & -\gamma + i\omega & 0 & -\varepsilon i \\ -\varepsilon i & 0 & -\gamma - i\omega & \varepsilon i \\ 0 & -\varepsilon i & \varepsilon i & -2\gamma \end{pmatrix}$$

Matriz de
densidad general
↑

$$\hat{\rho}_{ss} = \begin{pmatrix} 1-a & c \\ c^* & a \end{pmatrix} \Rightarrow \begin{pmatrix} 0 & \varepsilon i & -\varepsilon i & 2\gamma \\ \varepsilon i & -\gamma + i\omega & 0 & -\varepsilon i \\ -\varepsilon i & 0 & -\gamma - i\omega & \varepsilon i \\ 0 & -\varepsilon i & \varepsilon i & -2\gamma \end{pmatrix} \begin{pmatrix} 1-a \\ c \\ c^* \\ a \end{pmatrix} = \vec{0}$$

$$\begin{cases} c\varepsilon i - c^*\varepsilon i + a2\gamma = 0 & (1) \\ (1-a)\varepsilon i + c(-\gamma + i\omega) - a\varepsilon i = 0 & (2) \\ -(1-a)\varepsilon i - c^*(\gamma + i\omega) + a\varepsilon i = 0 & (3) \\ -c\varepsilon i + c^*\varepsilon i - a2\gamma = 0 & (4) \end{cases}$$

$$(1) = (4)$$

$$(2) + (3): c(-\gamma + i\omega) - c^*(\gamma + i\omega) = 0 \Rightarrow c = c^* \frac{\gamma + i\omega}{-\gamma + i\omega}$$

$$(1): a = \varepsilon i (c^* - c)$$

$$(2): c(-\gamma + i\omega) = \varepsilon i (2a - 1) \Rightarrow c = \frac{\varepsilon i (2a - 1)}{-\gamma + i\omega}$$

$$\text{Si } a =$$

Tamaño de ε

$$\varepsilon < \frac{\min_{n \neq m} |\lambda_n^{(0)} - \lambda_m^{(0)}|}{\|L_1\|} \rightarrow \Delta$$

$$\text{Para 1 TLS: } \lambda_n^{(0)} = 0, -\gamma + i\omega, -\gamma - i\omega, -2\gamma$$

$$\Delta = \min \{ \sqrt{\gamma^2 + \omega^2}, 2\gamma, 2\omega \} = 2\gamma$$

$$\|L_1\| = \max \{ |\lambda_{L_1}| \} = 2$$

$\hookrightarrow$ eigenvalores de L_1

$$\Rightarrow \varepsilon < \frac{0.2}{2} = 0.1$$

$$e^{4\gamma t} + 1 - 2 e^{-2\gamma t}$$